

On the Computational Complexity and Approximation of the General Routing Problem for Graphs in Nondiscretized Metric Spaces

Abstract

In this note, we analyze a simple upper bound on the additional cost incurred in the General Routing Problem (GRP) when the set of vertices and edges are required. We show that the GRP for nondiscretized metric graphs is NP-hard. We also show that augmenting the required edge set by connecting its components using minimal intercomponent distances yields a connected graph on which the Chinese Postman Problem (CPP) can be solved with a total augmentation cost that is at most triple the optimal augmentation cost of the original GRP instance, and at most twice for two connected component. This result is applicable in general metric graphs such as Euclidean geometric graphs.

1 Introduction

The General Routing Problem (GRP) is a generalization of the Chinese Postman Problem (CPP), in which a subset of vertices $Q \subseteq V$ and edges $R \subseteq E$ must be traversed in a minimum-cost closed walk. When where Q is disconnected with R , it is necessary to add additional edges to ensure the existence of an Eulerian tour. In this work, we provide a simple upper bound on the cost of this augmentation.

2 Preliminaries

Let (X, d) be a nondiscretized metric space if a metric space X with metric d satisfy the following.

Definition 1 (Nondiscretized Metric Space).

A metric space (X, d) is called nondiscretized if for any $a \in X$, and any $k > 0$, whenever $\exists b \in X$ s.t. $d(a, b) = k$, then $\forall r \in \mathbb{R}^+$, $\exists c \in X$ s.t. $d(a, c) = rk$.

This condition ensures that distances in the space can be scaled arbitrarily by natural numbers from any base distance.

For metric space (X, d) , let $G = (V, E)$ be a complete metric graph, where $V \subset X$ and $E = \binom{V}{2}$ with cost function as metric d . Let $Q \subseteq V$ and $R \subseteq E$ each denote the required vertex and edge set. Suppose the subgraph $G_R = (Q, R)$ contains $k \geq 1$ connected components:

$$G_R = C_1 \cup C_2 \cup \cdots \cup C_k.$$

Define:

$c(e)$: the cost of edge $e \in E$.

For $e = (v_1, v_2)$, $c(e) = d(v_1, v_2)$

$c(R)$: the total cost of edges in R , i.e. $c(R) = \sum_{e \in R} c(e)$.

T^* : additional edges needed in the optimal Eulerian solution of GRP for subgraph G_R

M^* : subset of T^* that connects all odd-degree vertices in G_R

H^* : subset of T^* that connects all disconnected components of G_R

$L(R)$: the minimum additional cost required to augment R into a closed Eulerian walk.
(i.e., the difference between the cost of an optimal GRP tour and $c(R)$); $L(R) = c(T^*)$

H : a minimum-cost set of edges connecting all components of G_R
(e.g., a minimum spanning tree over components).

M : a minimum-cost set of edges making all odd vertex of G_R have even degree
(e.g., a minimal matching over Euclidean perfect graph).

$J(R)$: the cost of a closed Eulerian walk obtained by an approximation algorithm,
specifically $J(R) = c(H' \cup M)$, where H' doubles all edges in H .

3 Main Result

3.1 NP-hardness

The Euclidean GRP is reduced to metric Traveling Salesman Problem (TSP).

Theorem 1. *The General Routing Problem in the metric space is NP-hard.*

pf. Consider the special case of the GRP where $Q = V$, $R = \emptyset$ for a complete metric graph $G = (V, \binom{V}{2})$. The subgraph G_R be finite set V of metric space (X, d) . Then the GRP reduces to finding a shortest Hamiltonian cycle where edges are allowed to appear twice.

Let $T^* = \bigcup_{i=1}^n \{(v_{i-1}, v_i)\}$ be an optimal solution corresponding to the shortest Hamiltonian cycle $(v_0, v_1, \dots, v_n = v_0)$. If some vertex v appears twice as $v_i = v_j = v$ for $i, j < n$. There exists adjacent edge $e_1 = (u, v)$, $e_2 = (v, w)$ so that $e_1, e_2 \in T^*$. Since the edges have cost as norm,

$$c(\{e_1, e_2\}) = c(e_1) + c(e_2) = d(u, v) + d(v, w) \geq d(u, w)$$

By replacing e_1, e_2 with $e = (u, w)$, new solution be $T = T^* \setminus \{e_1, e_2\} \cup \{e\}$. Thus,

$$c(T) = c(T^*) - c(\{e_1, e_2\}) + c(e) \leq c(T^*)$$

T corresponds to cycle that still pass through v , since v appears twice in Hamiltonian cycle. T forms a Hamiltonian cycle with shorter cost than the optimal solution T^* . By contradiction, optimal solution passes through all vertex exactly one time. The optimal solution is solution of Traveling Salesman Problem (TSP) on Euclidean plane.

We have to show that metric TSP is NP-complete.

metric TSP reduced to rectilinear Steiner tree algorithm, which is still NP-complete for metric space [1].

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□

If the metric space is nondiscretized, for nonempty R , or moreover G_R without trivial graph as connected component, it is NP-hard. We similarly show this by constructing G_R with no trivial graph as connected component, Euclidean GRP for G_R is reduced to metric TSP.

Theorem 2. *The General Routing Problem in nondiscretized metric space with nonempty R is also NP-hard.*

pf. Let $P = \{p_i\}_{i=1}^n \subseteq X$ be arbitrary. For finite P , find $s = \max_{i,j \in [1,n] \cap \mathbb{N}} d(p_i, p_j)$. Let's construct a disjoint graph C_i s as following.

$$C_i = (Q_i, R_i) = (\{p_i, p_{i1}, p_{i2}\}, \{(p_i, p_{i1}), (p_i, p_{i2}), (p_{i1}, p_{i2})\})$$

Take any j , then $d(p_i, p_j) = \alpha \leq s$. Since (X, d) is nondiscretized metric space, for some $k, l \in \mathbb{R}$, $\exists p_{i1}, p_{i2} \in X$ s.t.

$$(4i - 1)s \leq d(p_i, p_{i1}) = k\alpha \leq (4i)s \quad \text{and} \quad (4i - 1)s \leq d(p_i, p_{i2}) = l\alpha \leq (4i)s$$

For any point p_{i1} (or p_{i2}) find a lower bound of distance between disjoint group C_j .

$$\begin{aligned} d(p_{i1}, p_j) &\geq d(p_{i1}, p_i) - d(p_i, p_j) \\ &\geq (4i - 1)s - s \\ &= (4i - 2) \cdot s \geq 2s \end{aligned}$$

Also, for distance between p_{i1} (or p_{i2}) and p_{j1} (or p_{j2}) find a lower bound. WLOG let $i > j$

$$\begin{aligned} d(p_{i1}, p_{j1}) &\geq d(p_{i1}, p_i) - d(p_i, p_{j1}) \\ &\geq d(p_{i1}, p_i) - (d(p_i, p_j) + d(p_j, p_{j1})) \\ &\geq (4i - 1)s - s - (4j)s \\ &= (4i - 4j - 2)s \geq 2s \end{aligned}$$

Therefore, if $i \neq j$, ($\forall p \in Q_j$) we can say $d(p_{i1}, p) \geq 2s$.

Solve GRP for unions of C_i . Let $Q = \bigcup_{i=1}^n Q_i$, then take Euclidean perfect graph $G = (Q, \binom{Q}{2})$, and $R = \bigcup_{i=1}^n R_i$, then $G_R = (Q, R)$.

Each connected components C_i has even degrees for all vertex. For an optimal solution T^* , in general T^* must contain some set H^* that connects C_i s, and M^* that connects every odd degree vertices to ensure Eulerian connectivity. However, in this case, M^* is not needed, so $T^* = H^*$.

For any connection between C_i, C_j , we've shown the lower bound of cost between point p_{i1} (and p_{i2}) and C_j . Thus,

$$(\forall p \in Q_j) \quad d(p_{i1}, p) \geq 2s \geq d(p_i, p)$$

Taking edge connecting C_i, C_j , it is always optimal to choose p_i for vertex in C_i . Also, for C_j , taking p_j is optimal. Therefore $M^* \subset \binom{P}{2}$. Therefore, solving T^* is solving TSP in metric space for point P given in complete metric graph $G_P = (P, \binom{P}{2})$. Therefore, in case of nondiscretized metric space, for nonempty R Euclidean GRP is at least as hard as metric TSP, and is thus NP-hard. $\square$

3.2 Approximation Algorithm

Taking minimal matching of odd vertices in subgraph G_R and taking double edge obtained by Minimum Spanning Tree (MST) for connected components C_i s becomes an approximation algorithm.

Theorem 3. *The additional cost of approximation algorithm is at most triple the additional cost in the optimal GRP solution.*

$$J(R) \leq 3L(R)$$

More specifically, if G_R has only two connected components, the additional cost of approximation algorithm is at most double the additional cost in the optimal GRP solution.

pf. Let T^* be an optimal solution to the GRP instance on R . Then T^* must contain some set $H^* \subseteq E \setminus R$ including duplicates that connects all disjoint components of G_R , and $M^* \subseteq E \setminus R$ that connects every odd degree vertices to ensure Eulerian connectivity. Take all odd vertices of $G_R = (Q, R)$ so that the minimal matching M making all vertex degree even. There is a polynomial time algorithm of finding M [2] [3] [4]. Since M is a minimum-cost set of edges that connects all components, we have:

$$\text{cost}(M) \leq \text{cost}(M^*) \leq L(R).$$

Next, take minimum distance between disjoint C_i s. Let G_c be a complete graph $G_c = (\mathcal{C}, \binom{\mathcal{C}}{2})$, where $\mathcal{C} = \{C_i | i \leq k\}$. The cost of edge (C_i, C_j) be minimum distance between vertex in C_i and C_j .

Consider for case of $k = 2$.

Since adding T^* forms a Eulerian cycle crossing both C_1 and C_2 , we can give a direction. Take any vertex (WLOG in C_1) be starting point of the cycle. Following the (directed) Eulerian cycle, there exists a edge from $C_1 \rightarrow C_2$ and $C_2 \rightarrow C_1$, since Eulerian cycle contains a vertex in C_2 and end at starting point in C_1 . The edge added between two component for H^* is at least two. The minimum cost of (C_1, C_2) always smaller or equal to any of those edges. Let H'_2 be set of double edge for (C_1, C_2) , typically corresponds to some $v_1 \in C_1$ and $v_2 \in C_2$ s.t. $c(v_1, v_2) = c(C_1, C_2)$. Thus,

$$c(H'_2) = 2c(v_1, v_2) = 2c(C_1, C_2) \leq c(H^*) \leq L(R)$$

For general $k > 2$, we have to construct H in more general way.

A minimum spanning tree (MST) provides a polynomial-time method to connect all components of G_c with the smallest total cost [5] [6] [7]. Let H be the union of edges obtained by MST.

Generate H' by changing all edge in H into double edge. Adding double edge increases vertex degree of 2, so doesn't change the existence of Eulerian cycle. The additional edges $M \cup H'$ for G_R satisfy:

1. H' connect the component C_i s
2. M guarantees the all vertex be even degree

Therefore, $G' = (Q, R \cup M \cup H')$ is connected graph with Eulerian cycle.

Since T^* has H^* that connects every component C_i s, let L be set of all linked C_i, C_j by H^*

$$L = \{(i, j) | u \in C_i, v \in C_j \text{ for } (u, v) \in H^*\}$$

Then $\forall (u, v) \in H^*$, for $u \in C_i, v \in C_j$ $c((u, v)) = d(u, v) \geq d(C_i, C_j)$ by definition of $d(C_i, C_j)$. The total cost of H^* is bounded below

$$c(H^*) \geq \sum_{(i,j) \in L} c(C_i, C_j) \geq c(H)$$

The $c(H)$ is minimum cost for edges that connects all C_i, C_j , so one case of connecting $(i, j) \in L$, the cost of all connected edges are always larger than $c(H)$. Thus,

$$c(H') = 2c(H) \leq 2c(H^*) \leq 2L(R)$$

It follows that the matching cost $J(R) = c(R \cup M \cup H')$ has upper bound as following.

For $k = 2$,

$$J(R) - c(R) = c(M) + c(H') \leq L(R) + L(R) = 2L(R).$$

For $k > 2$,

$$J(R) - c(R) = c(M) + c(H') \leq L(R) + 2L(R) = 3L(R).$$

□

3.3 Special case

First, Show that Planar Straight line graph is also NP-hard!! Then, start the following ones. Need more thinkings.....

Since geometric graph is defined for metric space $(\mathbb{R}^n, d)$, where d is a L_t norm for $t \in \mathbb{N} \cup \{\infty\}$. We can say $(\mathbb{R}^n, d)$ is nondiscretized metric space. Then they satisfy the condition of previous results. Therefore, the approximation still holds for geometric graph, more specifically for planar graph.

In order to define planar straight line graph, consider the following relation and mapping in planar graph.

Let $\mathbb{R}^2 \times \binom{\mathbb{R}^2}{2} \rightarrow \mathbb{R}^2$ be the projection map ϕ for planar graphs. For any planar graph $G = (V, E)$, define the projection of G in $\mathbb{R}^2$ as:

$$\phi(G) = V \cup \{tv_1 + (1-t)v_2 \mid t \in [0, 1], (v_1, v_2) \in E\},$$

where each edge $(v_1, v_2) \in E$ is mapped to the set of points along the straight-line segment between the vertices v_1 and v_2 .

We define an equivalence relation $\sim$ on planar graphs as $G_1 \sim G_2 \iff \phi(G_1) = \phi(G_2)$. $G_1 \sim G_2$ means that they are equivalent in their projections on $\mathbb{R}^2$ i.e. their geometric embeddings coincide when viewed in the plane.

Now, we define a binary operation $*$ for combining two planar graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$. After combining two graph, it should remain planar, so generate vertex at every intersection between edges.

Let V_3 be the set of intersection points between the edges of G_1 and G_2 in $\mathbb{R}^2$:

$$V_3 = \{\phi(e_1) \cap \phi(e_2) \mid e_1 \in E_1, e_2 \in E_2, |\phi(e_1) \cap \phi(e_2)| = 1\}.$$

These are the intersection points of the edges e_1 of G_1 and e_2 of G_2 , where the intersection contains exactly one point. The set of vertices for the new graph $G = G_1 * G_2$ is:

$$V = V_1 \cup V_2 \cup V_3.$$

We also have to consider the case of edges intersects as line, and also generate new edges based on new vertex set V . For each edge $e = (v_1, v_2) \in E_1 \cup E_2$, we define the edge in G as a sequence of straight-line segments:

$$E(e) = \{(t_i v_1 + (1-t_i)v_2, t_{i+1}v_1 + (1-t_{i+1})v_2)\}_{i=1}^{n-1},$$

where $\{t_i\}_{i=1}^n$ is the set of parameters $t \in [0, 1]$ s.t. $t_i v_1 + (1-t_i)v_2$ is a point in V , and $0 = t_1 < t_2 < \dots < t_n = 1$. These parameters correspond to the vertices in the new graph formed by the intersection of edges of G_1 and G_2 .

The edge set E of the new graph G is the union of the edge sets of G_1 and G_2 , augmented with the new edges formed by the intersections:

$$E = \bigcup_{e \in E_1 \cup E_2} E(e).$$

It is obvious to say that the binary operation $*$ is associative, meaning that for any three planar graphs G_1, G_2, G_3 :

$$(G_1 * G_2) * G_3 = G_1 * (G_2 * G_3).$$

This follows from the fact that the edge intersection points are well-defined and the process of combining graphs via intersections does not depend on the order in which graphs are combined.

Also, the empty graph $G = (\emptyset, \emptyset)$ serves as the identity element for the operation $*$, as combining any graph with the empty graph results in the original graph:

$$G * (\emptyset, \emptyset) = G = (\emptyset, \emptyset) * G.$$

The collection of all planar graphs, denoted P , together with the binary operation $*$, forms a monoid $\langle P, * \rangle$.

Definition 2 (Planar Straight Line Graph (PSLG)).

*The planar straight line graph is the quotient monoid of $\langle P, * \rangle$ by the equivalence relation $\sim$.*

$$PSLG = P / \sim = \{[G] \mid G \in P\} \quad \text{where} \quad G_1 \sim G_2 \iff \phi(G_1) = \phi(G_2).$$

This quotient set of equivalence classes of graphs forms planar straight line graph, where the equivalence classes correspond to different geometric realizations of graphs in $\mathbb{R}^2$.

These are the distinct equivalence classes under the relation $\sim$, and they represent the geometric objects formed by projecting planar graphs onto the plane such that their embeddings are equivalent under homeomorphisms.

In this definition we can generally transform a geometric graph in $\mathbb{R}^2$ into planar straight line graph by generating vertex in all intersection of edges.

Consider a GRP in infinite complete graph, only conceptual meaning that any point in plane can be considered as vertex, $G = (V \simeq \mathbb{R}^2, E \simeq \mathbb{R}^4)$. The approximation algorithm still holds for planar straight line graph.

pf) We can still say that odd vertices should be even, so taking minimal matching of odd vertices still holds. $c(M) \leq c(M^*) \leq L(R)$.

Definition 3 (Triangular region).

For a graph G , triangular region of G is union of all triangles generated by edges in G .

Take $c(C_i, C_j)$ be distance between triangular region of C_i, C_j . Since we've generated planar straight line graph by adding vertex in intersection of edges, the disjoint C_i s are also disjoint in their triangular region. $c(C_i, C_j) > 0$. We can find vertex (maybe new) in $v \in C_i, u \in C_j$ s.t. $d(u, v) = c(C_i, C_j)$. Then adding u, v to each C_i, C_j are equivalent to each C_i, C_j , since u, v is at least in the line segment of edge in each C_i, C_j .

If we take H be edges selected by MST of C_i , we can still say $c(H') \leq 2c(H^*) \leq 2L(R)$.

Therefore, the approximation algorithm is still 3-approximation algorithm.

4 Discussion

We've shown that even by restricting GRP to a nondiscretized metric space still shows NP-hard. For the 3-approximation algorithm for metric space, this bound shows that a simple heuristic which first connects components using MST of minimum intercomponent distances and then solves the Chinese Postman Problem yields a tour whose cost is within a factor of 3 of the optimal GRP tour. This approach is especially relevant in geometric graphs, such as those in the Euclidean plane, where shortest Eulerian cycle can be computed efficiently.

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